

Interview Questions — Java - Web Technologies In Java

Total Questions: 44

1. What is the difference between a static and a dynamic web page?

Hint: Think about how the content is generated and whether it changes based on user input or server-side logic.

2. Explain the client-server model in the context of web applications.

3. How does HTTP communication work between a browser and a web server?

4. In a typical Java web application, describe the data flow from a user submitting a form on the frontend to the database storing the data.

Hint: Consider the roles of the frontend, servlet, and database.

5. What is a Java Servlet and why is it used in web development?

6. List and briefly describe the three main methods in the Servlet lifecycle.

7. What is the purpose of the web.xml deployment descriptor in a Java web application?

Hint: Think about configuration and how servlets are mapped.

8. How would you deploy and access a simple 'Hello Servlet' on Apache Tomcat?

9. What is the difference between the doGet() and doPost() methods in an HttpServlet?

Hint: Consider when each method is typically used.

10. How do you retrieve form parameters sent from a client to a servlet?

Hint: Recall the method from the HttpServletRequest object.

11. Write a simple code snippet in a servlet that reads a 'username' parameter from a request and prints it in the response.

12. Explain the difference between ServletConfig and ServletContext.

Hint: One is specific to a servlet, the other is shared across the application.

13. How would you store and access a database connection string as a context parameter in your web application?

14. What is the purpose of `sendRedirect()` in a servlet? How does it differ from `RequestDispatcher.forward()`?

Hint: Consider what happens to the client's browser and the URL.

15. Name and briefly describe the meaning of HTTP status codes 200, 302, and 404.

16. After a successful login, how would you redirect a user from a `LoginServlet` to a `DashboardServlet`?

17. What are cookies, and how are they used for session tracking in Java web applications?

18. Explain how the `HttpSession` API helps maintain user login state across multiple requests.

19. Describe one scenario where URL rewriting would be used for session tracking.

20. How would you keep a user logged in until they explicitly log out or the session expires? Briefly outline the steps.

21. What is JSP and how does it differ from a servlet?

Hint: Think about who writes each, and how dynamic content is generated.

22. Describe the JSP lifecycle in brief.

23. What is the purpose of JSP directives such as `%@ page %>`?

24. How would you use Expression Language (EL) in JSP to display a user's name stored in a request attribute?

25. Compare dynamic rendering in JSP with dynamic rendering in a frontend framework like React.

26. What does the `action` tag do in a JSP page?

Hint: Think about code reuse and including common layouts.

27. How would you use `<jsp:useBean>` and `<jsp:setProperty>` to work with JavaBeans in a JSP?

28. Explain the benefit of using custom tag libraries in JSP.

29. If you want to include a common header and footer in all your JSP pages, how would you achieve this?

30. Explain the Model-View-Controller (MVC) pattern as it applies to Java web applications using Servlets and JSP.

Hint: Describe the roles of Servlet, JSP, and JavaBean.

31. In a student registration application, which component would handle data validation and which would display the registration form?

32. How does data flow from a form submitted by the user to being displayed on a JSP page using the MVC pattern?

33. Why is server-side validation important, even if you have client-side validation in your forms?

34. Write a simple server-side check in a servlet to ensure a 'password' parameter is not empty.

35. How would you configure a custom error page in JSP to handle 404 errors?

36. Describe how you would display an 'Invalid Credentials' message to a user after a failed login attempt.

37. What steps are involved in connecting a Java Servlet to a MySQL database using JDBC?

38. Write a code snippet to fetch a list of students from a database and forward it to a JSP for display.

39. How can you display database records in a table format on a JSP page?

40. Explain how the MVC pattern helps when integrating JDBC with Servlets and JSP for database operations.

41. If you were to build a small web portal for student registration and event posting, what would be the main components and their responsibilities?

Hint: Think about controllers, models, views, and how data flows.

42. Describe the end-to-end flow of a user registering for an event in the Campus Connect Portal project.

43. How could you use an AI tool like ChatGPT or GitHub Copilot to help you debug a servlet or JSP issue, and how would you make sure the suggestions are safe to use?

Hint: Think about using AI for code help but also the need to test and review suggestions.

- 44.** How might AI-powered tools speed up your web development process in Java, and where would you still rely on your own skills?
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